

Article

(Doing) Computational History: The Role of Data Work in Computational Approaches

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Abstract

Computational methods have become increasingly prominent within the historical sciences, generating significant enthusiasm among some scholars. Yet their practical demands, epistemic limits, and ethical implications are less often critically examined than praised. This article explores what it means to *do* computational history today, arguing that it is not primarily defined by algorithms but by datasets. It is methodologically specific, resource-intensive, selective in scope, labour-heavy, and dependent on pre-digitised sources, specialised infrastructure, and interdisciplinary collaboration. These dependencies limit the scope of research questions and can produce narrow outcomes despite substantial effort, lending some validity to the concern over whether the field yields sufficient historiographical return for the labour invested. Corpus construction and data work lie at the epistemic core of computational history. These often undervalued tasks are not merely technical precursors to analysis, but interpretive and epistemic acts. Data are shaped by digitisation politics, historical bias, and institutional power. They shape the questions asked, the answers produced, and the legitimacy of findings. Recognising and valuing data work is essential, both to embed critical perspectives into computational humanities and to counteract the privileging of certain forms of labour over others. Due to the association of quantification with rigour and scholarly prowess, algorithmic work receives more credit, creating a two-tier system in this division of labour in which those who develop algorithms are elevated above those who curate data, despite their symbiotic interdependence. Computational history, when done well, requires deep engagement with our sources, be they historical or data. For computational history to stabilise as a meaningful discipline, it must prioritise building better datasets over pursuing increasingly complex algorithms on an unstable basis of data.

Keywords: computational history; digital humanities; computational humanities; data work; knowledge work; corpus criticism; data ethics



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1. Introduction

As computational methods become more prominent within the historical sciences, it is time not only to discuss their epistemic potential and technical implementation, but also to ask what it means to *do* computational history. What are the practical constraints of this work and what is the nature of the labour itself? This article seeks to address these questions not only by characterising computational history through its practices and requirements, but also by examining the specific constraints these pose: being resource-intensive, selective in scope, labour-heavy, and reliant on pre-digitised sources as a starting point. Computational history depends on specialised infrastructure and interdisciplinary

collaboration. These dependencies limit both the scope of possible research questions and the number of scholars able to engage in such work. Computational history also foregrounds practices such as corpus construction and data work. These are central epistemic activities, yet current research does not sufficiently acknowledge them as such. Much of this labour remains invisible, especially in contrast to the highly visible and highly valued outputs of algorithmic processes. Yet data work constitutes the epistemic foundation of computational history. This article outlines these dynamics and argues that, for computational history to establish itself as a meaningful discipline, it must prioritise building and curating high-quality datasets and valuing data work as an equal contribution.

Computational history emerged as a marginal pursuit within the historical discipline, drawing its methodological roots from fields such as historical statistics and the history of economics, which introduced quantitative techniques to historical research during the 1960s and 1970s (Aydelotte 1966). These early efforts were often confined to narrowly defined problems, and, already then, quantitative methods were frequently criticised as addressing only niche questions.

The broader adoption of computational methods gained momentum with the rise of digital humanities (Berry 2011; Roth 2019), particularly through the development of techniques such as distant reading (Binkyte 2023; Houston 2023; Jockers 2013; Moretti 2005, 2013; Primorac et al. 2023; Sinclair and Rockwell 2016)¹ and, more recently, distant viewing (Arnold and Tilton 2019, 2023).² The increasing stability and accessibility of machine learning technologies by the late 2010s made these tools more usable for humanities research. Although some scholars had already employed such methods earlier, this period marked a turning point in their institutional and intellectual visibility. The establishment of the Computational Humanities Research Conference in 2020 symbolises this shift: its founders sought a dedicated venue for computational work, feeling underserved by traditional conferences, although many have argued that such approaches had already been privileged within digital humanities circles (Lang 2020; Dombrowski 2022, p. 138). Nonetheless, the conference's rapid growth, alongside the expansion of AI-based methods, has solidified the prominence of computational approaches in the field. By 2023, the public and scholarly impact of large language models, particularly ChatGPT, dramatically increased interest in and awareness of computational methods among historians.³ Tools that had once appealed primarily to specialists were now being taken up by a broader cohort, including digital humanists who had previously worked according to more traditional, primarily qualitative paradigms.⁴ The accessibility of these models, as well as their role in facilitating academic publication, further propelled their integration into historical research. What had previously been a largely theoretical or peripheral interest has since moved into the centre of scholarly discourse. This is the backdrop against which the present position paper is situated.

As a position paper, this article foregrounds a meta-discussion of how computational history studies are conducted, offering insights that also apply more broadly to computational humanities projects. Having provided a brief history of computational history, the article now turns to its central question of what it means to *do* computational history, hence the title. It does not aim to establish definitive theses about computational history, nor does it engage in debates over disciplinary nomenclature—whether one prefers the terms computational history, digital history, digital humanities, or computational humanities (Roth 2019). It also does not attempt to evaluate the epistemic promises or methodological limitations of computational approaches, which have been discussed extensively elsewhere (Allington 2022; Bernhart 2018; Lang 2021; Lauer 2020; Piotrowski and Neuwirth 2020; Shadrova 2021). Instead, this article examines the practical dimensions of and labour involved in doing computational history. It outlines a set of characteristics that recur across

visible and influential projects in the field, irrespective of whether the practitioners identify as computational historians, digital historians, or simply as historians. These characteristics are drawn from observed practices and the author's own experience (Lang 2025), and some may even appear surprising from the perspective of traditionally trained historians. The aim is to articulate what it currently means to undertake computational historical work, and, in particular, to offer desiderata and directions for the future. This article begins by characterising computational methods as requiring a high degree of precision, which in turn shapes the kinds of projects that computational history and, by extension, computational humanities can realistically undertake. The discussion then moves to the resources required to run such projects, along with the corpus criticism and politics of digitisation that structure the field. The article then draws on a case study applying computer vision algorithms to early modern alchemical laboratory apparatus, illustrating the specific challenges involved in such work. While this case study represents only one project within a vast landscape of possibilities for computational history, it surfaces several structural and methodological issues that remain under-articulated. These include the labour of annotation, the limits of general-purpose tools when applied to historically specific materials, the necessity of interdisciplinary collaboration and the hidden infrastructures that support high-quality computational outputs. Bringing these elements to the fore is essential to understanding what it means to do computational history. It also serves to highlight the centrality of datasets to computational research and leads into a broader reflection on the ethical implications of a field that often creates a two-tier structure in its inevitable division of labor where algorithmic work is valued more highly than data work (Alvarado 2022; Lang 2027). Ultimately, the article argues that data work is pivotal to results derived from computational history work and must be recognised as such. It concludes that, despite the growing presence and accessibility of computational methods, this does not mean that everyone must adopt them. Rather, there remains significant potential for contributions from scholars with traditional historical skills, particularly in the context of data preparation and critical engagement with sources.

2. Precision and the Scope of Computational History

Computational historical research is typically conducted in highly specific contexts. It is most viable where substantial digitised source material already exists and where specialised teams are capable of leveraging the appropriate computational methods. The need for both digital infrastructure and methodological expertise means that such projects are, by their nature, selective in scope and resource-intensive in execution.

Much of this work is currently exploratory, testing the possibilities and limits of applying computational techniques to historical data. It is important to recognise that such methods do not lend themselves to broad generalisation. As in chemical analysis, computational tools can only detect what they are designed to measure; they cannot reveal phenomena outside the scope of their modelling (Stachowiak 1973). While computational approaches are often celebrated for identifying patterns otherwise inaccessible through traditional methods, their applicability in historical research remains less firmly proven.⁵ Often conducted by scholars in stable, well-funded environments—such as projects on modelling lost books using unseen species analysis (Kestemont et al. 2022), or the *Sphaera* project (Valleriani 2020, 2025a)—projects that yield genuinely novel insights tend to focus on very specific, granular questions, often requiring years of groundwork and significant labour to make any inroads.⁶ The cited successful and highly prominent examples demonstrate the potential of computational methods, but also highlight their infrastructural demands: identifying viable sources, acquiring or developing suitable tools, training models, and collaborating across disciplinary boundaries. These are not endeavours that can easily be

undertaken without significant institutional backing and long-term commitment. This exemplifies the broader point about precision: many computational projects are highly targeted and resource-intensive, producing focused results rather than sweeping generalisations. There is often a misconception that computational methods in the humanities require little effort and yield large-scale, generalisable insights. In practice, it is frequently the opposite: a great deal of work is required to produce results tailored to specific datasets or questions, and these methods rarely transfer seamlessly across different types of humanities data, which are typically complex and heterogeneous.

Given the specificity of computational techniques, research questions must be formulated with unusual precision. These methods typically require not only digitised sources in formats compatible with algorithmic analysis, but also a clear understanding of what can and cannot be measured. This may result in a binary or confirmatory structure of analysis, which is sometimes criticised as being in contrast with the interpretive ambiguity characteristic of much historical scholarship, resulting in more traditional historians sometimes assessing their outcomes as too small, niche or only marginal improvements on the state of scholarship despite the significant investments required to get them. However, one could argue that these criticisms pertain equally to more traditional forms of historical scholarship.⁷ Nevertheless, a defining feature of computational history is this dual nature: it is both labour-intensive and can still sometimes be narrow in outcome. Despite the visibility of a few high-profile projects, it is not wrong to be cautious about overstating their generalisability. Computational analysis may generate additional data points, but these still require qualitative interpretation. There is a fundamental distinction between the outcomes of a computational process and the historical interpretation drawn from them. If a model is poorly aligned with the historical phenomenon under investigation (Stachowiak 1973), it may yield little more than a technical analysis of the corpus itself (Lang 2021).

Another set of concerns related to computational methods more broadly centres on corpus construction, dataset documentation (Gebru et al. 2021), and model auditability (Brown et al. 2021; Vecchione et al. 2021). Questions of representativeness are particularly acute in historical research, where available data is uneven and shaped by prior processes of preservation and digitisation (Lang and Suárez Cronauer 2026; Zaagsma 2023). Because computational methods often rely on datasets shaped by these contingencies (Bender et al. 2021; Paullada et al. 2021), the resulting findings must be interpreted cautiously and with attention to their evidentiary limits.

While some argue that AI reduces labour by automating analysis, others have pointed out that it often shifts the burden rather than diminishing it (Crawford 2021). Historians may find themselves transformed from researchers into fact-checkers of AI outputs, or into technicians tasked with maintaining complex pipelines (Lang 2026). The time required for data preparation, computational modelling, and interpretive synthesis is considerable—particularly for scholars working outside formal project teams or lacking institutional support. Although individual scholars may undertake such work, it demands mastery of diverse technologies and disciplines, each of which could constitute a field in its own right. Such projects must begin by identifying suitable source material, preferably choosing material that is difficult to analyse through traditional means, and then selecting computational methods well-matched to the specific problem. They require collaborative effort and long-term investment, encompassing everything from data cleaning and project design to historical interpretation and publication.

Nevertheless, despite these constraints, the current phase of experimentation remains productive and worthwhile. Since we are not yet at a stage where large-scale or rapid computational history is feasible, carefully chosen case studies are valuable for assessing their potential. Importantly, these projects raise broader disciplinary questions: what

role should AI and computational methods play in the historical sciences? How can we distinguish between computational humanities work and more conventional digital history? And, crucially, will the results of computational history be recognised as meaningful contributions to historical scholarship, or dismissed as marginal experiments with limited interpretive reach? At present, funding and institutional interest remain strong, driven in part by the novelty and momentum of current AI developments. Yet, in the longer term, historians will need to assess whether the returns justify the costs—not only in financial and environmental terms, but also in scholarly labour and epistemic value.

3. Resources, Corpus Criticism, and the Politics of Digitisation

A central concern in computational history—frequently raised by traditionally trained historians—is whether such work yields insights that could not otherwise be achieved (Jannidis 2019).⁸ In response, this article argues that meaningful computational historical research must begin with a close and critical engagement with the available sources, particularly those already digitised. This reliance on digitised material is both a practical and conceptual necessity: without existing digital resources, the effort required to create usable data for computational analysis is immense and often beyond the reach of individual scholars or under-resourced institutions.

However, the digitisation of historical sources is shaped by longstanding dynamics of power, preservation, and access (Zaagsma 2023). Digitisation processes tend to reproduce historical biases, often privileging elite, canonical materials—typically white, male, and European—because these were among the first collections preserved and made available at scale (Dziudzia and Hall 2020; Hall 2019). As a result, the available digital record is far from neutral. Its composition reflects both past cultural and research priorities and present-day institutional decisions, which continue to influence what is studied and how.

Given this context, two criteria should guide source selection in computational history: first, the bulk of the sources under study should ideally already be digitised to make the endeavour feasible within typical project constraints, and second, they should either have been previously studied or be difficult to examine without computational methods. This ensures that the resource-intensive nature of digital research, including its environmental and infrastructural costs (Lang 2026), can be justified through demonstrable scholarly benefit. However, care also needs to be taken to mitigate the tendency to digitise and study only what is already well-known or canonised.

Corpus construction therefore emerges as a central epistemic and ethical concern. Like traditional historical source criticism, but intensified, corpus criticism requires researchers to interrogate who and what is represented in a dataset—and who is not (Lang and Suárez Cronauer 2026). It is here that computational history intersects most directly with conventional historical methodology. Constructing or selecting a corpus is not a neutral technical task but an interpretive act that shapes the questions that can be asked and the answers that can be given. As such, corpus criticism ought to be foregrounded as a core practice in computational history. Beyond describing, documenting and auditing existing datasets (Brown et al. 2021; Gebru et al. 2021; Vecchione et al. 2021), it will be the responsibility of Digital Humanists to help curate datasets useful for computer-assisted study. This not only involves concerns such as representativeness and balancedness of corpora but also actively digitising to fill in gaps in the record as part of this process. For example, this can be achieved by first auditing who is represented in a dataset and either enriching the dataset with extant information on underrepresented and marginalised groups that may only be tangentially represented in the existing source material or to add, for instance, 20% of new material of hitherto unrepresented sources and perspectives to combine with the 80% already digitised and well-prepared material.

Well-constructed and well-documented corpora are a prerequisite for sound computational research. This includes not only digitisation and metadata, but also the documentation of choices around inclusion, exclusion, and processing. In literary studies, for instance, some scholars have shifted from examining canonical texts to analysing patterns such as reprinting frequency, using corpus-based methods to explore popularity and cultural reception (Odebrecht et al. 2021; Schöch et al. 2021). Results of computational methods heavily depend on the quality and scope of the underlying data. Corpus work supports efforts to make historical narratives more inclusive. Without this foundation of knowing what's in one's data, even the most advanced computational models risk producing results that may be technically sophisticated but misrepresent historical societies. At the modelling stage, further epistemological challenges arise. A computational model must be carefully aligned with the phenomenon it aims to represent (Stachowiak 1973). If the alignment is weak, the model may yield insights only about the structure of the corpus itself, rather than the historical processes under study. While such work can contribute to identifying bias or absences in archives, it may fall short of making substantive historiographical interventions.

4. Case Study: Applying Computer Vision to Alchemical Images

In my own research, I undertook an experiment in applying computer vision techniques to the visual culture of early modern alchemy, with a particular focus on laboratory objects. The intention was to develop this as one chapter within a broader monograph, yet the practical challenges of doing so without dedicated funding or formal project infrastructure soon became apparent. A project of this complexity, interdisciplinary by nature and technically demanding, quickly revealed the limits of what a single researcher can realistically achieve, particularly when the specific computational work to be undertaken is not the central focus but only one component of a larger study. The project was conceived during what can retrospectively be described as the 'ChatGPT revolution' of 2023 when all sorts of computational methods suddenly seemed poised to become viable 'out of the box' due to widespread enthusiasm surrounding generative AI. However, the leap in AI capabilities was far greater in terms of public perception than in actual technological advancements (Narayanan and Kapoor 2024). Nonetheless, swayed by this optimism, I began early experiments using off-the-shelf computer vision tools, supported by a network of colleagues whose methodological expertise made the initial stages of the project possible (Lang et al. 2023). Object detection algorithms had to be trained to recognise unfamiliar, domain-specific visual features.⁹ These initial trials quickly revealed the limitations of the available tools; they are impressive as their web demo features may be seen, not because the algorithms themselves are less performant than advertised, but because early modern alchemical images diverge fundamentally from the kinds of visual materials used to train standard models.

In contrast, projects applying computer vision to 19th-century photographs or comparable, more modern visual sources have reported significantly more successful outcomes (Smits and Wevers 2023; Smits and Kestemont 2021; Wevers and Smits 2020). These results are not surprising as such images are visually more similar to contemporary photographs and tend to involve objects (such as humans, pets, or household items) that are well-represented in contemporary Internet data and widely used object detection classification schemes like MS-COCO (Lin et al. 2014). This demonstrates that existing computer vision algorithms, shaped by their underlying datasets and accordingly, the visual culture of the internet, support object detection tasks that align sufficiently well with the aesthetics and subject matter of modern life. This also means that attempting to identify a human, a cat, or even a zebra in a Victorian photograph is, in technical terms, a categorically different task from detecting an alchemical cucurbit or distillation apparatus in a 17th-century wood-

cut. However, our 2023 study (Lang et al. 2023), for instance, found that style transfer, i.e., having to bridge the visual difference between early modern etchings and modern-day photographs, was not a major issue. The challenge posed by early modern sources thus lies not only in their visual unfamiliarity to contemporary models but also in the absence of comparable concepts in the labelled data, causing a lack of conceptual understanding of these historical objects. As a result, many of the successful outcomes in existing historical computer vision projects may offer an unrealistically optimistic view of what is feasible for less conventional or canonical materials.

This absence of relevant objects within the training data proved to be one of the most significant constraints in detecting alchemical apparatus computationally and pertained to alchemical apparatus simply not existing in modern vision datasets. Today's computer vision models are built on surprisingly specific categories.¹⁰ A close examination of widely used training datasets such as MS-COCO (Common Objects in Context) reveals the over-representation of a narrow range of object categories, such as baseball bats and zebras, which results in their frequent appearance in classification outputs. Despite the name of the dataset, the objects represented in MS COCO are only 'common' within very specific historical and cultural contexts.¹¹ If one's Humanities data happens to align with these categories, standard object detection algorithms may perform reasonably well. Although MS COCO is just one of many datasets used to train object detection algorithms (Salari et al. 2022), some of which include a much larger range of categories, the broader conclusions still hold: the datasets on which computer vision models are trained are highly culturally and historically specific. As a result, these algorithms may fail to generalise when confronted with elements not represented in their training data.

Critical data studies (Iliadis and Russo 2016) have firmly established the outsized influence of datasets on the functioning of algorithms (Paullada et al. 2021). This has profound implications for computational humanities research. When faced with uncertainty, algorithms tend to default to categories they have encountered frequently in training. Thus, any vaguely elongated object may be labelled a 'baseball bat,' regardless of the utter irrelevance of that particular object category to the historical material at hand. This misclassification is not a failure of the model per se, but a reflection of the biases embedded in the underlying datasets. Such temporal and cultural biases become visible when computational methods are applied to source material that lies far outside the domain of contemporary visual culture. Other biases such as gender or race become particularly visible when models are applied to individuals who experience intersectional forms of marginalisation. This has been demonstrated vividly by projects such as *ImageNet Roulette* (Crawford and Paglen 2019) and by algorithm audits showing that certain computer vision systems performed worst on compounded forms of intersectional discrimination, with the largest performance gap observed between white men and Black women, due to the absence of relevant training data (Buolamwini and Gebru 2018; Raji et al. 2020).

More often than not, biases in datasets only come into focus years after their creation, typically when the dataset is applied outside the context for which it was originally developed and begins to show performance issues. Even the widely used ImageNet dataset, which in its original form contained dehumanising, sexist, and racist labels, was used extensively across a broad range of domains and tasks for over a decade (Luccioni and Crawford 2024). It was only when datasets came under public scrutiny during the 2020 Black Lives Matter movement that these issues were more widely acknowledged and addressed. In the context and aftermath of the movement, many popular datasets were publicly called out for serious flaws (Birhane et al. 2021a, 2021b, 2024; Rajabi et al. 2022). This led to the withdrawal of the *80 Million Tiny Images* dataset (Torralba et al. 2008) and prompted cleanup efforts in others, such as ImageNet (Denton et al. 2021; Yang et al. 2020).

Among the most unexpectedly fundamental aspects in my own project was the manual labor of annotation. I was struck by how little attention is given to data work (Alvarado 2022) in the broader discourse surrounding computational methods, especially given the outsized influence of annotation quality and conceptual clarity of selected labels on the final outputs. Annotation is not merely a prerequisite for the model pipeline but an epistemically central research process in its own right. It compelled me to engage with the source material repeatedly and provoked reflection on our use of classification categories adapted from previous work (Frietsch 2017). Matching the results of a linguistic project on alchemical terminology (Gaede 2024), I discovered that the names of alchemical apparatus often depended more on function than on visual form (Lang 2025). This insight emerged not from applying algorithms to our data, but only after reviewing hundreds of images, in ways not unlike traditional historical source analysis. This reinforced the broader point that annotation is not merely a preparatory step but an epistemic process that profoundly shapes the research itself. Yet, the genre conventions and space constraints of computational humanities publications rarely acknowledge this. In our first paper presenting the preliminary experiments (Lang et al. 2023), the methods section could only offer a cursory overview of our approach. Even within the team, reconstructing the many small, cumulative decisions made over time proved difficult in retrospect. Much of this work remained tacit, undocumented, and unpublished. Many computational humanities papers are very detailed when it comes to information regarding algorithms used while omitting the complex, iterative histories of data itself and the preparation that made the computational modelling possible in the first place. This omission has consequences, not only for appropriate scholarly credit, but also for transparency, reproducibility, and interpretability. Although it is increasingly common to publish code and datasets in open repositories, the interpretive decisions embedded in annotation often remain invisible.¹² Without thorough documentation, other researchers cannot fully understand or responsibly reuse the data. This represents a serious obstacle for the field, particularly given its ambitions for methodological rigour and reproducibility. Given the priorities of computational humanities publications, which often show little to no interest in annotation and data practices, deeming them irrelevant to computational work, if not in theory, so in practice by pushing them out of their conferences and publication venues, it may come as a surprise to new practitioners just how profoundly annotation decisions shape the outcomes of any computational work. This divide between computational work, where only algorithms tend to be valued, and the ‘merely digital’ work of data curation and annotation is one of the key issues this article seeks to challenge. Only by ignoring the obvious and well-known influence of data quality on algorithmic results can this divide be maintained. It is a genuine loss for the quality of our research that computational and digital humanities approaches so often remain siloed from one another.

Harkening back to our case study, despite my initial attempts to handle my project alone, I soon found myself reliant on the contributions of colleagues with complementary skills. A metaphor offered by one of my collaborators, Vojtěch Kaše, illustrates this well. He likens traditional historical scholarship to a concert pianist performing solo, scientific research to a symphonic orchestra led by a conductor, and digital humanities to a jazz band: a collective in which each participant brings a distinct expertise, takes the lead at different moments, and contributes improvisations to the whole. This strikes me as a fitting characterisation of the collaborative ethos required in computational humanities that resists the myth of the solitary researcher and acknowledges the distributed labour underpinning such work.¹³

We are beginning to see tangible progress in the recognition of alchemical objects, largely as a result of gradual methodological refinement, the laborious creation of dedicated

training data (Lang 2025), and the continued development of computer vision algorithms. However, these advances came only after years of trial and error in what was, by necessity, an intermittent and collaboration-dependent research process. This experience brought into sharp focus how rarely the sheer volume and complexity of the labour involved, especially the data work, is acknowledged, either in public discussions or in scholarly publications. Much of the discourse surrounding computational research continues to foreground improving algorithms, while downplaying or entirely omitting the extensive and time-consuming preparatory work that makes such research possible in the first place. In my own case, I have repeatedly questioned whether the results ultimately justified the immense labour involved, especially in light of the fragility and limited success of early experiments. Computational projects of this kind are far more demanding than they may appear from the outside, and the types of information and results presented in publications typically reflect only a small fraction of the work required to produce them. High-quality datasets, often developed over many years before the computational work begins, form the foundational layer of such projects. Yet this labour, though indispensable to achieve the final results, is rarely foregrounded in publications or talks. Without well-constructed, carefully curated data, computational work in the humanities simply cannot deliver meaningful results, no matter how sophisticated the algorithms may be.

5. A Two-Tier Society: Ethics and the Value of Labour in Computational History

The data-centric AI (DCAI) movement argues that ethical improvements lie in improving datasets rather than applying technical fixes to models (Jakubik et al. 2024; Jarrahi et al. 2023). In the context of historical research, where data is more often than not partial, biased, or inherently shaped by processes of exclusion, moving towards more ethics-forward scholarship will require significant labour, for example, in cleaning, contextualising, supplementing, and documenting data. This is not work that can be automated or rushed. It demands both scholarly expertise and sustained investment, and it must be recognised as such by funding bodies, institutional review boards, and research communities. However, this labour is unequally distributed and often undervalued (Lang 2027). Within both history and computer science, data work (Alvarado 2022) has historically been framed as secondary, “menial,” or feminised (Ross and Pilsch 2022; Nyhan 2023, pp. 134–35), and this pattern persists in computational humanities. Tasks such as digitisation, metadata creation, and corpus construction are essential to any computational project, yet are rarely foregrounded in publications or rewarded in academic hiring. Moreover, the highly valued and better compensated forms of (computational-forward) labour are not equally accessible: only those at well-resourced institutions, with significant training or the option to engage with it, or those operating within established research networks that allow them to draw on others to compensate for any lacks in resources, specialisation or training can typically engage in large-scale computational projects. This raises serious questions about equity and access in the emerging landscape of data-driven history.

The field’s growing valorisation of quantification contributes to these dynamics. Computational humanities is often perceived as more “rigorous” or “scientific” than traditional humanities work (Riley 2017)—a perception rooted in long-standing, patriarchal assumptions about the objectivity of numbers (Boyles 2018, p. 94).¹⁴ Yet, the drive towards quantification predates digital methods (Aronova et al. 2017) and has always been selective in its epistemic ambitions (Aydelotte 1966). While quantitative approaches can be particularly useful in fields such as economic history, where structured data enables insights not otherwise possible, this does not imply that all historical research should adopt data-driven methods. Nor does the presence of numerical outputs guarantee epistemological

value. Quantification is not an inherent marker of quality; it must be appropriate to the research question.

This becomes particularly important when considering what types of topics are granted visibility and legitimacy within computational spaces. Feminised topics or approaches—those grounded in feminist theory, cultural critique, or interpretive analysis—often receive less attention at major digital humanities venues. As the ADHO conference landscape has shown, even when gender parity is achieved in participation, topics associated with feminised intellectual labour tend to be undervalued (Eichmann-Kalwara et al. 2018). This imbalance reflects deeper structural hierarchies in how (intellectual) work is recognised and rewarded.

In this context, calls to move away from model-driven AI and towards data-driven AI gain particular significance. Rather than continually optimising models developed by commercial entities under opaque and ethically questionable conditions, computational historians might achieve greater impact by focusing on building transparent, reusable, and well-documented datasets (cf. Valleriani 2025b).¹⁵ While large language models may retain a role in specific applications, much can still be accomplished with more traditional machine learning techniques (Neudecker 2023), provided the underlying data is robust.

Importantly, the commonly cited lack of “ground truth” data for specific historical use cases that may drive many to ill-curated foundational models claiming to need little to no additional training data should not be accepted unquestioned. It points instead to a need for renewed attention to collaborative, even computationally supported data curation and enrichment to avoid over-reliance and dependency on models created by bad actors. Such efforts not only enhance the quality and inclusivity of research, but also shift value back to the interpretive and methodological core of the humanities. As the Digital Humanities community and its practices show, data work (Alvarado 2022) is not a departure from humanities practice, but a continuation of it. The curating, contextualising, and critical interrogation of data is continuous with the interpretive labour that underpins all historical scholarship. In this light, practices like distant reading or distant viewing are instructive not just for the insights they offer but for the ways they frame the relationship between method and interpretation. They underscore the continued and ever-present necessity of Humanities skills and critical judgment even in computational contexts.

Accordingly, the ethical dimensions of computational history, particularly in relation to AI and data-driven research, cannot be reduced to compliance or “ethics washing.” Instead, ethical reflection must be embedded within research design and implementation, supported by institutional infrastructures that enable scholars to engage with these challenges meaningfully. Existing guidelines, including checklists and frameworks, for example, developed as part of Data Feminism (D’Ignazio and Klein 2020; Klein and D’Ignazio 2024; Lang and Suárez Cronauer 2026) or in the broader AI ethics community, offer a starting point, but they are no substitute for substantive critical reflection. However, implementing ethical standards in computational research is resource-intensive. What is needed is a combination of awareness, funding, and practical tools that translate ethical concerns into actionable practices within historical research workflows.

Ethics in computational history, then, is not only about responsible use of AI. It is also about labour, representation, and recognition. It is about resisting the implicit hierarchies that devalue certain forms of work or marginalise certain voices. And it is about building systems—technical, scholarly, and institutional—that empower researchers to engage in practices that promote more ethical engagement.

6. Data Work in Computational History

A persistent misconception among more traditional historians is that computational historians are distant from their sources, relying solely on machine outputs and failing to engage with the actual source they claim to investigate. In practice, however, no matter what terms like “distant reading” suggest, computational projects often involve extensive hands-on work with one’s sources, both in preparing the data and in interpreting the results (Lang 2025).¹⁶ This labour is often invisible in final publications but constitutes a substantial part of the scholarly process. Contrary to popular assumptions, robust computational history is not merely a matter of applying algorithms to existing data. It is fundamentally grounded in the quality of the data itself, and in the labour-intensive, interpretive work that goes into preparing and understanding that data. While oversimplified applications of computational methods do exist and rightly attract criticism (Da 2019a, 2019b, 2020), well-executed computational history requires methodological expertise, careful corpus construction, and deep engagement with historical sources.

This article strongly argues for the recognition of data work as an essential and intellectually demanding component of computational history. The outcomes of any machine learning model or statistical method are only as sound as the datasets on which they rely.¹⁷ Historians and data practitioners, including librarians, metadata specialists, and digitisation teams, are indispensable in this process. Their labour, though frequently undervalued in the context of the interventions more closely associated with computer science, provides the empirical foundation for any computational analysis. Structured data is a prerequisite for computational work, and its creation is itself a specialised task. As such, the field will continue to rely on a division of labour: between those who produce data, those who analyse it computationally, and those who pursue interpretive methods without digital tools. These are distinct but complementary skill sets, and they should not be hierarchically ordered. Unfortunately, current academic dynamics often do just that.

The undervaluation of data work is not unique to the humanities. In computer science, the exploitative treatment of data workers—particularly in the Global South—has become well documented: These individuals are often underpaid and subjected to poor working conditions while performing the crucial labour of annotating and curating training datasets (Gray and Suri 2019). Although digital humanities projects may not directly engage in such practices, they too rely on invisible labour: book scanners, digitisation technicians, librarians, and bibliographic experts whose work enables computational research are rarely acknowledged (Nyhan 2023; Ross and Pilsch 2022).¹⁸ This raises broader ethical concerns. If we are to uphold principles of fairness and scholarly integrity, we must also ensure that the labour underlying computational work—especially that which is gendered, racialised, or geographically marginalised—is made visible and properly credited. Solidarity across disciplines and roles is essential, particularly given the hierarchical structures that tend to privilege model development over data preparation. Contemporary research cultures often celebrate the development of new models, particularly in the context of large language models (LLMs), while overlooking the more foundational challenge of building high-quality datasets following the spirit of data-centric AI (DCAI).¹⁹ This emphasis on model-driven AI comes at the expense of data-driven approaches, which may be more sustainable, transparent, and better aligned with the epistemic goals of the humanities. Rather than continually chasing marginal improvements in proprietary models which are often produced under opaque and ethically dubious conditions, we should prioritise the collaborative creation of trustworthy, well-documented datasets (Valleriani 2025b).

Data documentation (Gebru et al. 2021) is now receiving increased attention in both computer science and digital humanities, particularly in relation to reproducibility. While standards such as FAIR and CARE provide useful frameworks, they are not sufficient

without detailed metadata about dataset construction: what was included, what was excluded, and why.²⁰ Without such information, datasets cannot be meaningfully reused or critically assessed. There is now a wide range of methods for documenting datasets and models, including model cards for models (Mitchell et al. 2019), datasheets for datasets (Gebru et al. 2021), datasheets for digital cultural heritage datasets (Alkemade et al. 2023), and data envelopes (Luthra and Eskevich 2024). In addition, dedicated data paper formats have emerged, for example, in the *Journal of Open Humanities Data* (JOHD) or the *Zeitschrift für digitale Geisteswissenschaften* (ZfdG), among others. GLAM institutions play an important yet underacknowledged role in this space.²¹ Auditing can uncover some of this information retrospectively (Brown et al. 2021; Vecchione et al. 2021), but thorough documentation from the outset remains indispensable. Nonetheless, this work is under-credited within academic reward systems. Therefore, we must also reconsider the division of labour in digital projects. More historians are already entering digital humanities through data-focused roles, bringing with them the domain expertise necessary for high-quality annotation and corpus construction. This movement should be supported by valuing their expertise, not leaving them marginalised.

7. Conclusions

The perception that computational methods are more rigorous (Riley 2017) or ‘scientific’ has led to them becoming privileged in certain institutional settings, such as Digital Humanities conferences and funding structures, where technical contributions are more easily accepted than work grounded in cultural theory, hermeneutics, or feminist scholarship (Eichmann-Kalwara et al. 2018).²² This raises serious concerns about the direction of the field. If digital history becomes aligned too closely with a culture of programming and quantitative bias, it risks marginalising work that addresses questions of gender, race, class, and power—especially when these are explored through more traditional interpretive or qualitative frameworks. This dynamic reflects broader patterns of epistemic inequality, whereby quantitative methods are often seen as more legitimate simply because they deal in numbers. But quantification is not inherently valuable. There is no scholarly gain in measuring what cannot meaningfully be measured.²³

Many digital corpora reflect the priorities of early digitisation initiatives, which tended to reinforce established canons by focusing on elite, Eurocentric texts. As a result, many voices remain absent from the digital record despite being preserved in physical archives. Enriching corpora by integrating underrepresented perspectives is therefore an ethical and historiographical imperative, enabling more inclusive computational research.

Large-scale computational projects frequently depend on digitised sources that are already OCR-processed and pre-cleaned. Because most computational history projects must rely on pre-digitised materials—with digitisation and pre-processing being time-intensive and frequently still underestimated, thus holding the potential to massively derail research project timelines—computational research frequently begins not with a traditional historical question, but with a triangulation of available sources, methodological constraints, and domain knowledge. This demands a different workflow from conventional historical research, in which a question is typically formulated first and then pursued through source discovery. Preparatory steps, such as digitisation, cleaning, and metadata creation, consume significant time and can delay or derail research if underestimated. Some digital humanists have begun using computational tools to assist with these tasks, but much of the work remains manual and interpretive. Corpus curation and design will thus have to become a key scholarly activity: identifying existing sources, and hopefully supplementing them with marginalised perspectives in striving for greater representativeness.

Among traditional historians, the popular misconception persists that computational approaches are inherently decontextualising, distancing researchers from their sources. Yet in practice, careful Humanities-driven machine learning research involves extensive, hands-on interaction with primary materials (Lang 2025). Far from replacing historical expertise, computational history depends on it—particularly in the phases of data selection, annotation, and interpretation. Curating better datasets will increasingly become a key area of Digital Humanities work. Such curation is essential for computational history to be truly useful to the wider historical discipline. While there is value in experimenting with cutting-edge methods, such as LLMs, many of these experiments produce highly specific results of limited relevance to broader historiographical concerns (Lang 2021). While specificity is not inherently problematic (indeed, it is a scholarly virtue), overinvestment in niche technical outputs risks alienating the broader community of historians. Focusing on well-curated datasets in the spirit of data-centric AI offers an alternative path by supporting more grounded, reproducible, and inclusive research.

Despite this, it is unlikely that all historians will—or should—become data-driven researchers or historian data workers. Ultimately, not all research should be—or can be—data-driven. The current dominance of computational methods must be tempered by critical awareness of their limitations, their politics, and the uneven access they presuppose. For computational history to develop responsibly, it must remain open to multiple methodological traditions and attentive to the power dynamics embedded in its data, its tools, and its institutional frameworks. The skills required for computational history differ from those in traditional historiography, and there will always be a need for scholars whose work focuses on data creation and contextualisation. In fact, this diversity of expertise is necessary for the field to function. Yet, disciplinary and institutional structures often fail to value these contributions equally. Nevertheless it is important to highlight that practices encompassed under the umbrella of data work are neither external, foreign nor new to the humanities; they are, in large part, a continuation of their interpretive and methodological traditions (Alvarado 2022). Distant reading and distant viewing, as two prominent examples of computational methods, deliver data that must still be interpreted within the framework of historical argument. Computational research thus retains, at its core, the epistemic practices of the humanities: engaging critically with historical sources (now especially in the form of corpora, rather than microhistorical examples), contextualising them, and interpreting the information available, whether it results from traditional or computational research. Historical knowledge is essential in evaluating what kinds of sources are available, how they can be interpreted, and what questions are appropriate to ask.

For computational history to flourish as a legitimate and inclusive scholarly practice, we must recognise the centrality of data work, value the labour that underpins it, and shift attention away from computational aspects alone. By building better datasets, curating more inclusive corpora, and documenting our sources with care, we lay the groundwork for meaningful collaborations between historians, data workers, and computational scholars. Only then can computational history move beyond niche applications and make broader contributions to historical knowledge.

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Abbreviations

The following abbreviations are used in this manuscript:

DH	Digital Humanities
LLM	Large Language Model
ADHO	Alliance of Digital Humanities Organizations
GIGO	Garbage In, Garbage Out
GLAM	Galleries, Libraries, Archives, and Museums
DCAI	data-centric AI
OCR	Optical Character Recognition
FAIR	Findable, Accessible, Interoperable, Reusable
CARE	Collective Benefit, Authority to control, Responsibility, Ethics

Notes

- ¹ Distant reading denotes a widely used approach in the digital humanities that employs computational techniques to analyse textual corpora. The analysis is conducted ‘at a distance,’ often through quantitative or statistical procedures that abstract from individual passages and instead operate at the level of large datasets. While the term is closely associated with Franco Moretti, whose work catalysed its adoption, contemporary practice has diverged considerably from his original formulation, and his fairly black-and-white approach has been subject to sustained critique. It is therefore important not to conflate the distant reading turn in Digital Humanities, its practitioners and methods, with the relatively narrow concept Moretti proposed when he first introduced the term over two decades ago (Moretti 2005, 2013; Primorac et al. 2023). In practice, the majority of digital humanists do not construe distant and close reading as mutually exclusive alternatives. Rather, they adopt mixed-methods hybrid reading approaches that combine computational analysis with more traditional forms of text interpretation, operating at the intersection of distant and close reading (Aledavood 2024). It would therefore be misleading to suggest that practitioners of distant reading do not consult their texts directly, or that they regard engagement with their source materials as unnecessary. In fact, close reading is usually needed to make sense of distant reading results. Accordingly, Matthew Jockers, for example, has reframed Moretti’s original proposal as a process of ‘zooming in and out,’ arguing that the term *macroanalysis* more accurately describes what computational text analysis can accomplish (Jockers 2013). These methods, of course, do not ‘read’ in a human sense; rather, they analyse text and detect patterns across large corpora, which must then be interpreted within broader historical and literary frameworks. It is also important for those unfamiliar with computational humanities to recognise that distant reading now encompasses a wide range of methods that extend far beyond what was possible when the paradigm first emerged. Early computational text analysis often relied on relatively simple techniques, such as counting word frequencies (Sinclair and Rockwell 2016), but the available methods have since diversified considerably. However, due to the slow diffusion of Digital Humanities methods back into their Humanities disciplines of origin, scholars who are new to or unfamiliar with the field often still associate distant reading with these early, sometimes rather naive approaches (e.g., word clouds). Beyond word frequency analysis, methods most commonly used today include, for example, clustering, topic modelling, sentiment analysis, and many more. The field is broad, with significant variation in how methods are implemented and their underlying assumptions. Not all methods are equally effective in every context: for instance, topic modelling is particularly useful for large corpora with diverse themes, but less so for small or homogeneous datasets. Sentiment analysis is also common, although it can be problematic methodologically, as it relies on predefined sentiment dictionaries that can yield divergent results depending on which dictionary is used. Many of these methods are used in digital history (Lässig 2021), adapted to serve historical inquiry, though methodological developments usually originate in computational literary studies. Ultimately, they draw on natural language processing and linguistics but are repurposed for humanistic questions that differ from the concerns of those source disciplines.
- ² Distant viewing (Arnold and Tilton 2019, 2023) as a research paradigm, coined in response to distant reading, emerged from the critique that digital humanities have been overly text-focused, neglecting other media (Binkyte 2023). In response, scholars have called for a visual (Wevers and Smits 2020) or multimodal (Smits and Wevers 2023) turn in computational humanities. The term ‘distant viewing’ refers to the use of computer vision methods to analyse visual cultural artefacts. While common applications such as optical character recognition (OCR), handwritten text recognition (HTR) or, to use the more general term, Automated Text Recognition (ATR) fall under the broader umbrella of computer vision (Hodel 2023), distant viewing typically involves

more complex image analysis, such as identifying and comparing features across large image corpora. Examples include the study of image reuse or the reuse of print plates (Dutta et al. 2021; Götzelmann 2022), combining concerns of document analysis and layout recognition. Distant viewing is also applied in film analysis or photographic collections (Arnold and Tilton 2023). These methods are often used by libraries or cultural heritage institutions for large-scale tasks like enhancing recommendation algorithms. However, adapting them for scholarly research questions requires a high degree of precision and effort, as the methods do not generalise easily on materials radically different from those on which they have been trained.

It is important to note that the perceived AI revolution is largely due to increased public awareness and a greater appreciation of AI as a useful tool, rather than any sudden technological leap. Providing a historical perspective that traces AI's gradual development over the past 80 years, the authors of the excellent book *AI Snake Oil* (Narayanan and Kapoor 2024) challenge the popular narrative that AI is on the brink of a singularity. They argue that the excitement surrounding tools like ChatGPT in 2023 was driven more by increased visibility, public awareness and public perception of current AI technologies than by the magnitude of genuine breakthroughs. In their view, the widespread perception of an ongoing AI revolution reflects shifting public attention rather than the scale of actual technological advancement—although this heightened interest has certainly led to drastically increased scholarly attention and an onslaught of AI-related papers across all disciplines, which may now actually accelerate progress. The surge of AI-related publications across disciplines further reinforces the impression of a revolution, even if the underlying developments remain incremental.

This surge of work on generative AI has prompted many institutions in the humanities, digital humanities and beyond, to publish statements on the use of AI in research. These statements often focus specifically on generative AI, but also address AI more broadly and the implications of its development in corporate contexts, which is discussed in another contribution to this special issue. One widely circulated example is the manifesto *Against the Uncritical Adoption of 'AI' Technologies in Academia* (Guest et al. 2025).

For example, claims that “the sum of the data points is greater than their parts” in analyzing social media data to make previously unseen patterns visible (Lasser 2023) are frequently invoked in computational social sciences.

The *Sphaera* project, led by Matteo Valeriani, is a long-term study of multiple editions of the same book, examining how these editions changed over time and were modified by different printers (Valleriani 2020, 2025a). To support this work, the project developed a specialised database based on an extended CIDOC-CRM ontology to represent relationships not typically captured in standard bibliographic metadata. In addition, computer vision analyses were conducted on diagrams within these editions. While the project produced valuable insights, one could argue that, despite its computational methods often associated with large-scale analysis, it centres on a single book, not unlike a scholarly edition project would. However, in computational contexts, expectations frequently lean towards a broader scope, even when such expectations conflict with the precision and level of detail required to obtain meaningful results using computational methods. The project on lost books applies extinct species algorithms from biology in a cross-disciplinary transfer of methods, aiming to estimate how many books may have been lost over time (Kestemont et al. 2022). While the approach is certainly innovative and the large-scale results are headline-worthy, the method is highly specialised and demands significant effort to adapt. The outcome, though intriguing, remains speculative, as there is no way to verify the estimates. Given these constraints, more traditional historians could question how much this contributes to historical understanding.

Following this line of argument, historical interpretation itself could be described using the metaphor of the black box often applied to algorithms (Schwandt 2022). This raises the question of whether computational approaches differ as fundamentally from traditional historical methods as is often assumed. This could be argued in many respects, ranging from their detail focus to their validity or explainability.

There has been a famous related debate including a considerable aftermath (Da 2019a, 2019b, 2020; Jannidis 2020; Underwood 2020; Ries et al. 2023; Joyeux-Prunel 2024).

Object detection involves two key steps: locating an object within an image and then labelling or classifying it correctly. While models can readily identify common objects like plants, animals, or humans (i.e., concepts well represented in training data), recognising and classifying specialised historical items is significantly more difficult.

A critical analysis of six widely cited benchmark datasets (Caltech 101, Caltech 256, PASCAL VOC, ImageNet, MS COCO, and Google Open Images) demonstrates how the creators' subjective choices and the labour of crowd workers shape the datasets: The selection of categories is not grounded in a general notion of visibility but is instead driven by perceived practical applications and the availability of downloadable images (Smits and Wevers 2021). Moreover, the reliance on Flickr and the broader web for data collection has introduced a temporal bias into many computer vision datasets.

The 80 object categories can be explored at <https://cocodataset.org/#explore>, accessed on 10 November 2025 (Lin et al. 2014).

However, it is important to note that even within the highly technical computational humanities community, where one might expect open-source code sharing to be standard practice, there remains considerable room for improvement. A recent study (Illmer 2025) found that all necessary code was cited in only 40% of the publications examined, and notably, this assessment did not even involve actually attempting to run the code. It merely checked whether the code was theoretically available for

replication. This highlights significant shortcomings in current practices and underscores the need for greater transparency and reproducibility in the field. In addition, other documentation and reporting best practices, such as datasheets, data and model audits, or carbon reporting (Lang et al. 2025), have not even entered the broader consciousness of the computational humanities field as measures that should be taken for ensuring transparency.

- 13 The myth of the lone male genius is unfortunately prevalent in many disciplines, including in Digital Humanities (Nyhan 2022).
- 14 The myth of the ‘objective algorithm’ has been debunked many times. For example, see (Crawford 2021).
- 15 On dataset criticism, see (Paullada et al. 2021; Orr and Crawford 2024) or on critical data studies: (Iliadis and Russo 2016).
- 16 Many digital humanists regret that terms like *Distant Reading* became so widely adopted that replacing them with terms more accurately reflecting what computational methods actually do has proven difficult. In fact, few digital humanists would endorse the original claims associated with the term as proposed by its inventor, Franco Moretti (Moretti 2005, 2013). Unfortunately, the critical receptions and subsequent reinterpretations of Distant reading within the field are often less visible to those outside of digital humanities. As a result, many hold unrealistic assumptions about what digital humanities scholars mean by the term. One attempt to introduce a more suitable alternative is Jockers’ *Macroanalysis* (Jockers 2013).
- 17 Cf. the “Garbage In, Garbage Out” (GIGO) principle.
- 18 For instance, the Contributor Role Taxonomy (CRediT, Holcombe 2019) has been proposed to democratize the attribution of credit beyond just the authors of academic papers.
- 19 There has been a lot of highly visible LLM criticism (Bender et al. 2021).
- 20 The CARE principles (GIDA 2021) are related to the increasingly more frequent call for an ethics of care (Gray and Witt 2021).
- 21 This issue is currently being debated within the digital humanities community in discussions on BlueSky sparked by Matthew Wilkens’ claim that digital humanities may have effectively ended, with much of what once fell under its umbrella absorbed into quantitative disciplines (Wilkens 2026). The claim generated considerable debate online: while some agreed, many strongly disagreed. Notably, a GLAM professional pointed out that practitioners in adjacent fields such as digital archives often feel excluded from discussions about the “state of the field,” although even narrow definitions of computational approaches to humanities data accurately describe their work. (<https://bsky.app/profile/did:plc:f4obdtap2xdezbn73lyo5dlu/post/3mf3cnwi4s22k>, accessed on 10 November 2025). This, again, reflects an unequal division and valuation of labour. Those who align themselves more closely with computer science or quantitatively oriented disciplines tend to receive greater visibility and recognition than those whose work remains rooted in more traditional humanities contexts or in forms of labour historically coded as feminised (Lang 2027), such as much of the work carried out in GLAM institutions. This structural imbalance within the field warrants our critical attention.
- 22 Research in digital humanities has shown that even in contexts where gender participation is balanced, topics coded as feminine are systematically less recognised. At ADHO conferences, for example, feminised themes often receive less visibility despite equal scholarly merit (Eichmann-Kalwara et al. 2018).
- 23 There is ample disciplinary discourse on the role of quantitative methods (Allington 2022; Bernhart 2018; Lang 2021; Lauer 2020; Piotrowski and Neuwirth 2020; Shadrova 2021).

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